

--	--	--	--	--	--	--	--

CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2016 Trial Examination

FORM VI

MATHEMATICS EXTENSION 2

Tuesday 9th August 2016

General Instructions

- Reading time — 5 minutes
- Writing time — 3 hours
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 100 Marks

- All questions may be attempted.

Section I — 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II — 90 Marks

- Questions 11–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 72 boys

Examiner

DNW

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The value of i^{2016} is:

- (A) i
- (B) -1
- (C) $-i$
- (D) 1

QUESTION TWO

At time t a particle is at position x and travelling with velocity v . Which of the following expressions is NOT equal to the acceleration of the particle?

- (A) $\frac{d^2x}{dt^2}$
- (B) $\frac{dv}{dt}$
- (C) $\frac{d}{dx}\left(\frac{1}{2}v^2\right)$
- (D) $\left(\frac{dx}{dt}\right)^2$

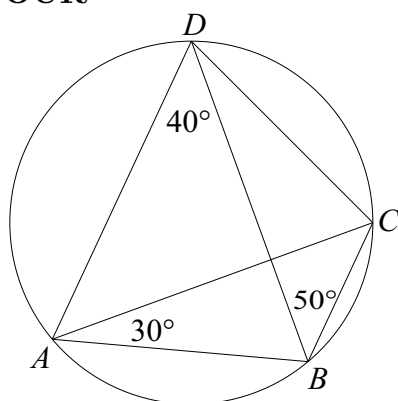
QUESTION THREE

Which expression is equal to $\int \frac{1}{x^2 + 2x + 3} dx$?

- (A) $\tan^{-1}\left(\frac{x+1}{2}\right) + C$
- (B) $\tan^{-1}\left(\frac{x+1}{\sqrt{2}}\right) + C$
- (C) $\frac{1}{2} \times \tan^{-1}\left(\frac{x+1}{2}\right) + C$
- (D) $\frac{1}{\sqrt{2}} \times \tan^{-1}\left(\frac{x+1}{\sqrt{2}}\right) + C$

Examination continues next page ...

QUESTION FOUR

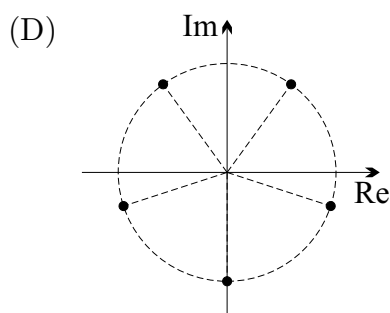
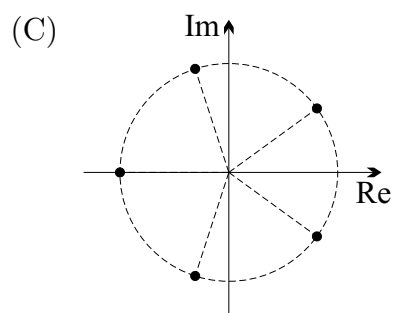
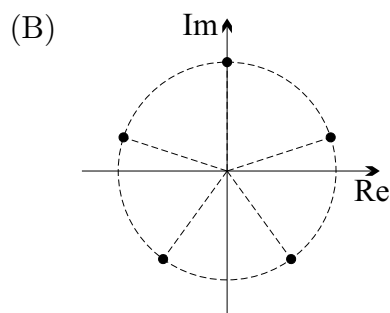
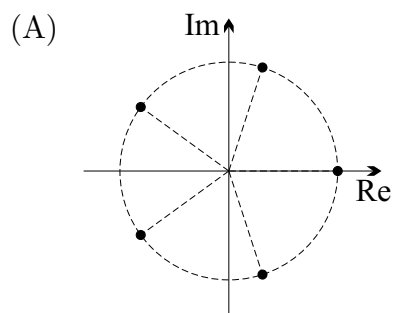


The sketch above shows cyclic quadrilateral $ABCD$ with $\angle BAC = 30^\circ$, $\angle ADB = 40^\circ$ and $\angle CBD = 50^\circ$. From this information, the size of $\angle ABD$ is:

- (A) 20°
- (B) 40°
- (C) 60°
- (D) 65°

QUESTION FIVE

Which of the following Argand diagrams shows the solutions of $z^5 - i = 0$?



QUESTION SIX

A certain function $f(x)$ has the following properties: $f(0) = 1$ and $\lim_{x \rightarrow \infty} f(x) = 3$.

Which of the following is possible for all values of x ?

- (A) $f''(x) > 0$ and $f'(x) > 0$
- (B) $f''(x) > 0$ and $f'(x) < 0$
- (C) $f''(x) < 0$ and $f'(x) > 0$
- (D) $f''(x) < 0$ and $f'(x) < 0$

QUESTION SEVEN

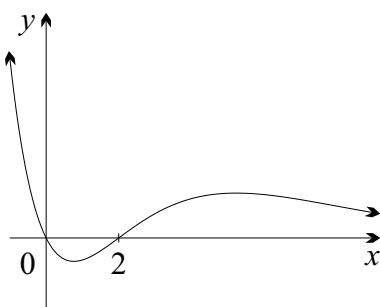
The polynomial $P(z)$ has real coefficients and $P(0) = -1$. The imaginary number α and the real number β satisfy:

$$P(\alpha) = 0, \quad P(\beta) = 0 \quad \text{and} \quad P'(\beta) = 0.$$

The degree of $P(z)$ is at least:

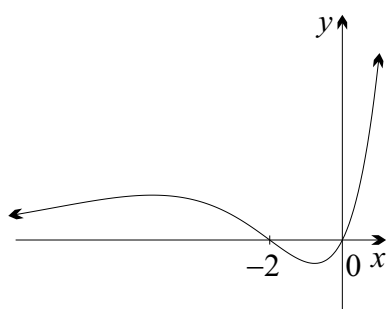
- (A) 2
- (B) 3
- (C) 4
- (D) 5

QUESTION EIGHT

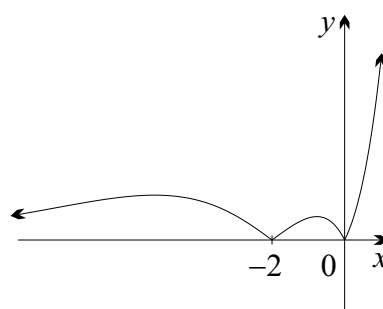


Above is the graph of $y = f(x)$. The correct graph of $y = |f(-x)|$ is:

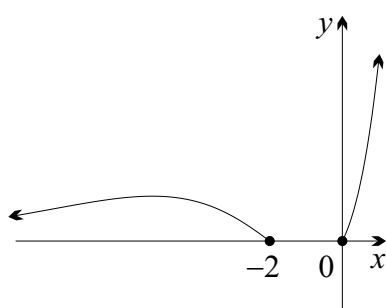
(A)



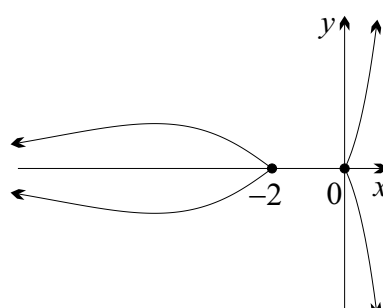
(B)



(C)



(D)



QUESTION NINE

Let $z = \cos \theta + i \sin \theta$, where θ is acute. The value of $\arg(z^2 + 1)$ is:

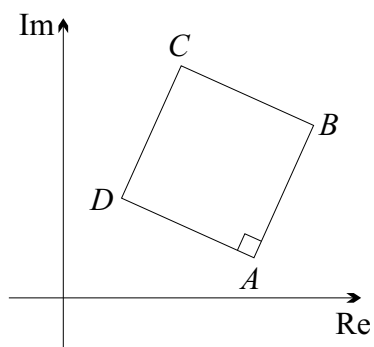
(A) $\frac{\theta}{2}$

(B) θ

(C) 2θ

(D) 4θ

QUESTION TEN



The Argand diagram above shows the square $ABCD$ in the first quadrant. The point A represents the complex number z and the point C represents w . Which of the following represents the point B ?

- (A) $\frac{z + w}{2} + \frac{i(z - w)}{2}$
- (B) $\frac{z + w}{2} - \frac{i(z - w)}{2}$
- (C) $\frac{z - w}{2} + \frac{i(z + w)}{2}$
- (D) $\frac{z - w}{2} - \frac{i(z + w)}{2}$

_____ End of Section I _____

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.**Marks**

(a) Express $\frac{5-i}{2+i}$ in the form $x+iy$, where x and y are real. 2

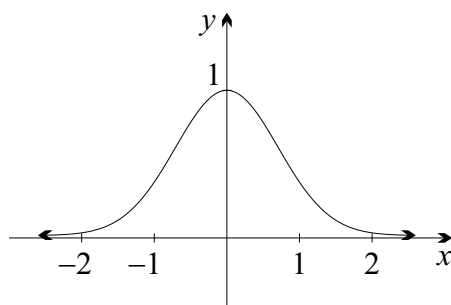
(b) Consider the region in the Argand diagram where $|z-2i| \leq 1$ and $\operatorname{Re}(z) \leq 0$.

(i) Sketch the region. 2

(ii) What range of values does $\arg(z)$ take? 2

(c) Use the substitution $t = \tan\left(\frac{x}{2}\right)$ to help evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin x + \cos x} dx$. 4

(d) 2



The graph of $y = e^{-x^2}$ is shown above. Draw a one-third page sketch of the graph of

$$y = (1-x^2)e^{-x^2},$$

without the aid of calculus. Show all intercepts.

(e) The region in the first quadrant below $y = 4x - x^3$ is rotated about the y -axis to form a solid. Use the method of cylindrical shells to find the volume of this solid. 3

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

(a) The equation $\frac{x^2}{4} - \frac{y^2}{5} = 1$ represents a hyperbola. Find the eccentricity e . **1**

(b) Find the gradient of the tangent to the curve $x^2 - 2xy + 3y^2 = 11$ at the point $(2, -1)$. **3**

(c) Let $P(z) = z^4 + 2z^3 + 3z^2 + 4z + 2$.
 (i) Show that $z = -1$ is a double root of $P(z) = 0$. **1**

(ii) The other two roots are complex. Use the sum and product of roots to find them. **3**

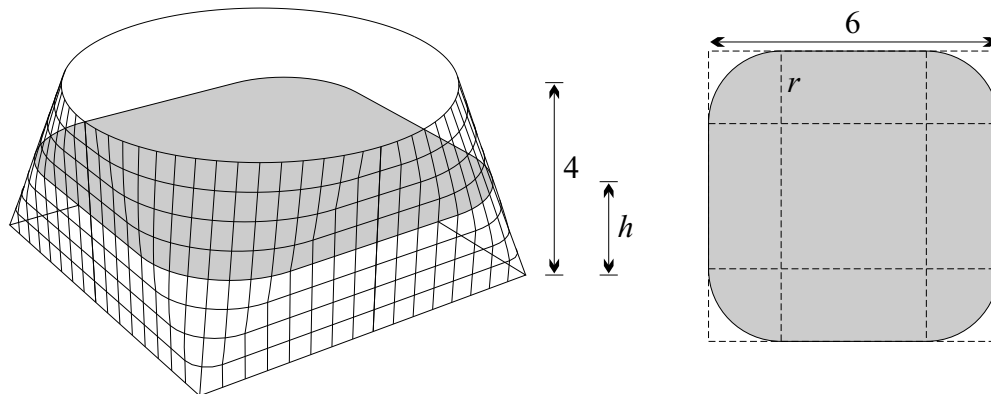
(d) A 1 kg object is moving along the x -axis and is subject to a resistive force of $\frac{1}{3}v^3 + v$, where v is its velocity in metres per second. That is, its equation of motion is **3**

$$v \frac{dv}{dx} = -\frac{1}{3}v^3 - v.$$

Initially the object is at the origin, and its velocity is 3 m/s.

Find v as a function of x .

(e)



An artist has created a sculpture which has a square base with side length 6 cm and a circular top with radius 3 cm. The height of the solid is 4 cm. A cross-section at height h is shown shaded grey, and a separate top view of the cross-section is shown on the right. This cross-section is a square with the corners replaced by quadrants of radius r . The radius r varies in direct proportion with h . That is, $r = kh$ for some constant k .

(i) Show that the area A of the cross-section is given by **2**

$$A = 36 + \frac{9}{16}(\pi - 4)h^2.$$

(ii) Hence find the exact volume of the solid. **2**

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.

Marks

- (a) (i) Suppose that $y = f(x)$ has a stationary point at $x = a$, and that both $f(a) \neq 0$ and $f''(a) \neq 0$. Let $g(x) = \frac{1}{f(x)}$.

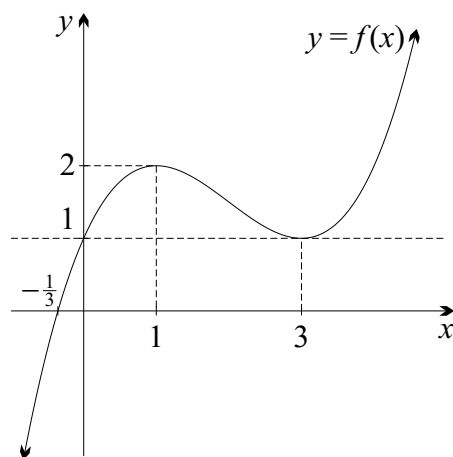
(α) Prove that the graph of $y = g(x)$ also has a stationary point at $x = a$.

1

(β) Show that at the stationary point the sign of $g''(a)$ is opposite to $f''(a)$.

2

(ii)

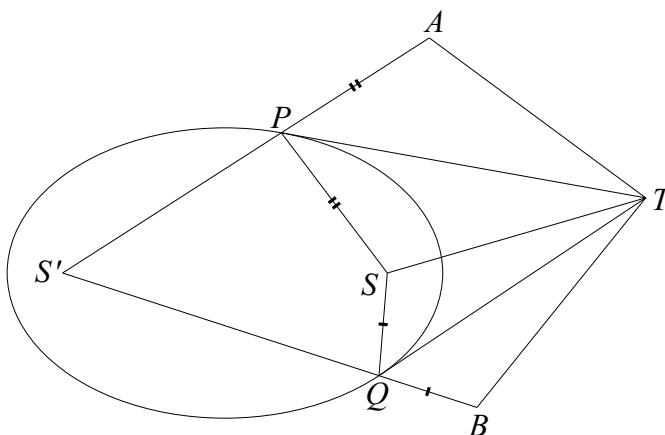


2

The graph of $y = f(x)$ is shown above. You may assume that both $f''(1) \neq 0$ and $f''(3) \neq 0$.

Use the results of part (i) to help sketch the graph of $y = \frac{1}{f(x)}$. Clearly show the behaviour at any turning points, and any other significant features.

(b)



The diagram above shows an ellipse with foci at S and S' . Let the length of the major axis be $2a$. The tangents from points P and Q on the ellipse meet at T . Extend $S'P$ to A with $PA = PS$, and extend $S'Q$ to B with $QB = QS$.

In this question you may assume the reflection property of the ellipse.

(i) Prove that $AT = ST$.

2

(ii) Explain why $S'A = 2a$.

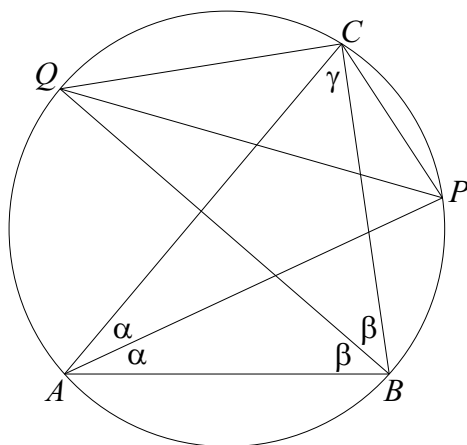
1

(iii) Hence prove that $S'T$ bisects $\angle PS'Q$.

2

Examination continues overleaf ...

(c)



The diagram above shows a circle with a chord AB of fixed length. A variable point C lies on the major arc AB . The angle bisector of $\angle BAC$ meets the circle again at P , and the angle bisector of $\angle ABC$ meets the circle at Q . Let $\angle BAP = \alpha$, $\angle ABQ = \beta$ and $\angle ACB = \gamma$.

Copy or trace the diagram into your writing booklet.

- (i) Show that $\angle PCQ = \alpha + \beta + \gamma$. 2
- (ii) Hence prove that $\angle PCQ$ is constant, regardless of the location of point C . 2
- (iii) Give a reason why the length of PQ is constant. 1

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.

Marks

- (a) Let $z = \cos \theta + i \sin \theta$. Then, by de Moivre's theorem, it is known that 3

$$z^n + z^{-n} = 2 \cos n\theta \quad \text{and} \quad z^n - z^{-n} = 2i \sin n\theta.$$

Use these two results to show that

$$16 \cos^3 \theta \sin^2 \theta = 2 \cos \theta - \cos 3\theta - \cos 5\theta.$$

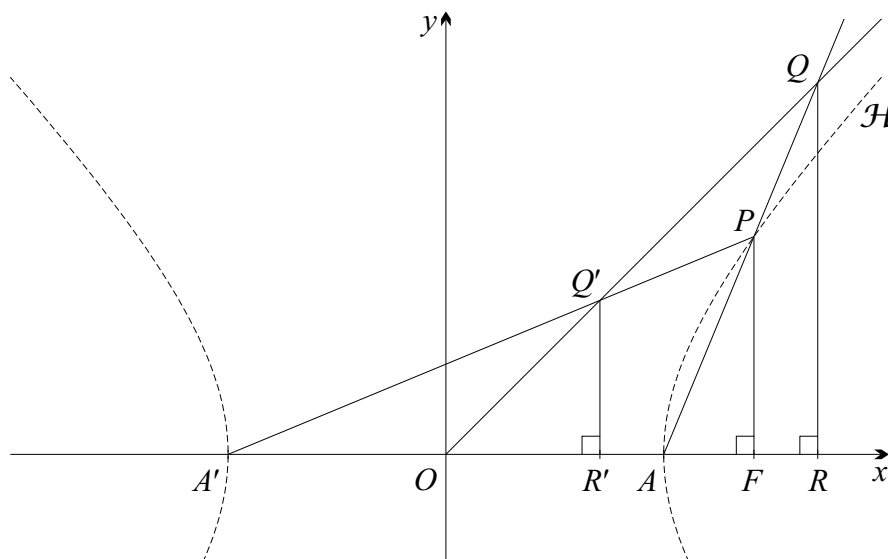
- (b) (i) Solve $z^5 = 1$. 1
- (ii) Let α be the complex root in the first quadrant of the Argand diagram. Show that $\alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = 0$. 1
- (iii) Find a quadratic equation which has roots $(\alpha^4 + \alpha)$ and $(\alpha^3 + \alpha^2)$. 2
- (iv) Solve this equation and hence evaluate $\cos \frac{2\pi}{5}$. 2
- (c) The polynomial $P(z) = z^4 - 2z^3 + 2z^2 - 10z + 25$ has two complex zeroes α and $i\alpha$.
 - (i) By considering the equations $P(z) = 0$ and $P(iz) = 0$, show that α is a root of $(1 + i)z^2 - 2z + 5(1 - i) = 0$. 2
 - (ii) Solve the above quadratic and thus find all the zeroes of $P(z)$. 4

Examination continues next page ...

QUESTION FIFTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) Find the locus of z given that $\frac{z-1}{z+1}$ is imaginary. 3

(b)



The diagram above shows the hyperbola $\mathcal{H}$ with vertices at $A(a, 0)$ and $A'(-a, 0)$, and centre $O(0, 0)$. The point $P(a \sec \theta, b \tan \theta)$ is in the first quadrant and on $\mathcal{H}$. Let F be the foot of the perpendicular from P to the x -axis. The asymptote $y = \frac{b}{a}x$ intersects AP at $Q(a\mu, b\mu)$ and intersects $A'P$ at $Q'(a\lambda, b\lambda)$. You may assume $\mu > \lambda$. The perpendiculars from Q and Q' meet the x -axis at R and R' respectively.

- (i) Show that $Q'Q = (\mu - \lambda)\sqrt{a^2 + b^2}$. 1
- (ii) You may assume that $\triangle AFP \parallel \triangle ARQ$. Use the ratios of matching sides to express $\frac{\mu}{\mu - 1}$ in terms of θ alone. 1
- (iii) Find a similar ratio for λ . 1
- (iv) Multiply your results for parts (ii) and (iii) and hence show that the length of $Q'Q$ is constant. 2

(c) By considering $(x - \frac{1}{x})^2$, show that $a + \frac{1}{a} \geq 2$ whenever $a > 0$. 1

(d) A 1 kg mass is projected vertically upward from ground level with initial velocity V_0 . As it moves it is influenced by gravity $g \text{ m/s}^2$ and an air resistance equal to kv , where v is its velocity in metres per second and k is a positive constant. Let y be its height above ground level after t seconds, with upwards as positive. You may assume that the equation of motion is

$$\ddot{y} = -g - kv.$$

(i) Show that the velocity at time t is given by 3

$$v(t) = \frac{g}{k} \left(\left(1 + \frac{k}{g} V_0 \right) e^{-kt} - 1 \right).$$

(ii) Show that the time T taken to reach the maximum height satisfies 1

$$e^{kT} = 1 + \frac{k}{g} V_0.$$

(iii) From part (i), the velocity s seconds before it reaches the maximum height is 2

$$v(T - s) = \frac{g}{k} \left(\left(1 + \frac{k}{g} V_0 \right) e^{-k(T-s)} - 1 \right).$$

It can be shown that $v(T + s)$ is the velocity s seconds after it reaches the maximum height, and that $v(T + s)$ is negative to indicate downwards velocity. (Do NOT show this.)

Use part (c) to help show that $v(T - s) + v(T + s) \geq 0$.

QUESTION SIXTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) (i) Find the values of
- A
- ,
- B
- and
- C
- such that

2

$$\frac{1}{x^3 + x^2 + x + 1} = \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 1}.$$

- (ii) Find and simplify
- $I(N) = \int_0^N \frac{dx}{x^3 + x^2 + x + 1}$
- .

3

- (iii) Hence evaluate
- $\int_0^\infty \frac{dx}{x^3 + x^2 + x + 1}$
- .

1

- (b) Let
- $I_n = \int_0^1 x^n e^{-x} dx$
- , where
- $n \geq 0$
- is an integer.

- (i) Use integration by parts to show that
- $I_n = \frac{1}{n+1} (e^{-1} + I_{n+1})$
- .

1

- (ii) Use induction to show that

3

$$I_0 = e^{-1} \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} \right) + \frac{1}{n!} I_n \quad \text{for } n \geq 1.$$

- (iii) You may assume that
- $\lim_{n \rightarrow \infty} I_n = 0$
- and that the series in part (ii) converges to a limit. Use these assumptions to show that

1

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots$$

- (iv) Suppose that
- e
- is rational. That is, suppose that
- $e = \frac{p}{q}$
- , where
- p
- and
- q
- are integers with
- $p > 0$
- and
- $q \geq 2$
- . Use part (iii) to show that

1

$$p(q-1)! = \left(\sum_{k=0}^q \frac{q!}{k!} \right) + \frac{1}{q+1} + \frac{1}{(q+1)(q+2)} + \dots$$

- (v) Let
- $f = \frac{1}{q+1} + \frac{1}{(q+1)(q+2)} + \frac{1}{(q+1)(q+2)(q+3)} + \dots$

2By comparing f with a suitable geometric progression, show that $0 < f < \frac{1}{2}$.

- (vi) Hence prove that
- e
- is in fact irrational.

1

 End of Section II

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

B L A N K P A G E

Multiple Choice (with comments on errors)

Q 1 (D) $2016 = 4 \times 504 + 0$ so $i^{2016} = i^0 = 1$.

(A) $i^{2017} = i$ (B) $i^{2018} = -1$ (C) $i^{2019} = -i$

Q 2 (D) This is v^2 , and $v^2 \neq a$.

(A) acceleration (B) acceleration (C) acceleration

Q 3 (D) $x^2 + 2x + 3 = (x + 1)^2 + (\sqrt{2})^2$ then use the formula for $\tan^{-1}$, $a = \sqrt{2}$

(A) a wrong, wrong formula (B) wrong formula (C) a wrong

Q 4 (C) $\angle CAD = 50^\circ$ (angle in same segment)

$\angle DBA = 60^\circ$ (angle sum of $\triangle ABD$)

(A) $\angle BDC = 30^\circ$ but opposite angles of $ABCD$ are NOT equal.

(B) Incorrect angle sum of triangle

(D) The diagram is not to scale.

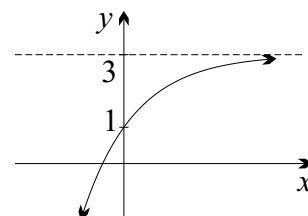
Q 5 (B) One solution must be $z = i$. The others are equally spaced from this.

(A) $z^5 - 1 = 0$ (C) $z^5 + 1 = 0$ (D) $z^5 + i = 0$

Q 6 (C) $f(0) < \lim_{x \rightarrow \infty} f(x)$ so $f'(x) > 0$

thus $f'(x) > 0$ and $\lim_{x \rightarrow \infty} f'(x) = 0$ so $f''(x) < 0$

(See the example graph on the right.)



The other options are wrong because:

(A) $f''(x) > 0$ (B) $f''(x) > 0$ and $f'(x) < 0$ (D) $f'(x) < 0$

Q 7 (C) β is a double root, and for real coefficients complex roots come in pairs.

(A) not just α and β

(B) omitted $\bar{\alpha}$ or omitted β as double root

(D) $P(0) = -1$ does not imply an extra zero

Q 8 (B) reflect in y axis, $y = |f| > 0$, no restriction on x .

(A) $y < 0$ (C) restriction on x (D) restriction on $x, y < 0$

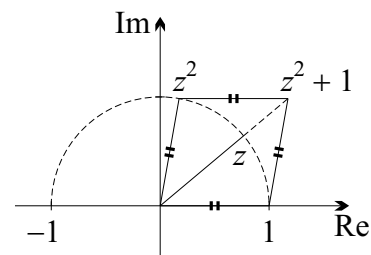
Q 9 (B) angles in a rhombus

Alternatively $z^2 + 1 = (\cos 2\theta + 1) + i \sin 2\theta$

$= 2 \cos^2 \theta + 2i \sin \theta \cos \theta$

$= 2 \cos \theta (\cos \theta + i \sin \theta)$.

(A) $\arg(z + 1)$ (C) $\arg(z^2)$ (D) $\arg(z^4)$



Q 10 (A) Let M be the mid-point of AC then $\frac{z + w}{2} = \overrightarrow{OM}$ and $\frac{z - w}{2} = \frac{1}{2} \overrightarrow{CA}$.

So $\overrightarrow{OB} = \overrightarrow{OM} + \overrightarrow{MB} = \overrightarrow{OM} + i \times \frac{1}{2} \overrightarrow{CA}$

(B) point D (C) $i \times \overrightarrow{OD}$ (D) $-i \times \overrightarrow{OB}$

QUESTION ELEVEN (15 marks)

Marks

- (a) Realise the denominator using its conjugate:

$$\frac{5-i}{2+i} = \frac{5-i}{2+i} \times \frac{2-i}{2-i}$$

$$= \frac{9-7i}{5} \quad \text{or} \quad \frac{9}{5} - \frac{7}{5}i$$

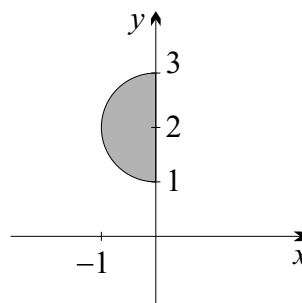
2

✓

✓

- (b) (i)

Inside and on the circle centre $2i$, radius 1 and in the left half-plane.



2

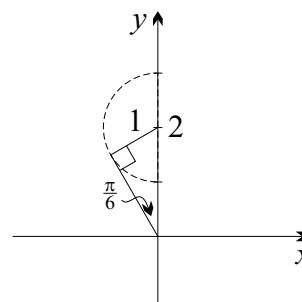
✓

✓

- (ii)

$$\frac{\pi}{2} \leq \arg(z) \leq \frac{\pi}{2} + \frac{\pi}{6}$$

so $\frac{\pi}{2} \leq \arg(z) \leq \frac{2\pi}{3}$



2

✓

✓

- (c) From $t = \tan \frac{\theta}{2}$

$$d\theta = \frac{2}{1+t^2} dt.$$

at $\theta = 0, \quad t = 0,$

at $\theta = \frac{\pi}{2}, \quad t = 1$

$$\begin{aligned} \text{so } \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta + \cos \theta} d\theta &= \int_0^1 \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \times \frac{2}{1+t^2} dt \\ &= \int_0^1 \frac{1}{(1+t)} dt \\ &= [\log(1+t)]_0^1 \\ &= \log 2 \end{aligned}$$

4

✓

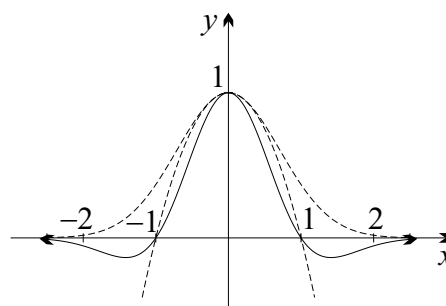
✓

✓

✓

- (d)

Intercepts shown at $(0, 1)$, $(-1, 0)$ and $(1, 0)$.
Even and horizontal asymptote shown.



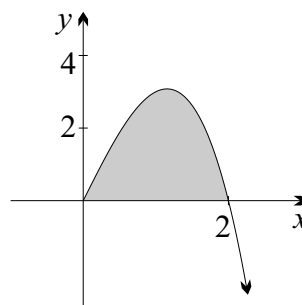
2

✓

✓

(e)

$$\begin{aligned}
 V &= 2\pi \int_0^2 x \times y \, dx \\
 &= 2\pi \int_0^2 4x^2 - x^4 \, dx \\
 &= 2\pi \left[\frac{4x^3}{3} - \frac{x^5}{5} \right]_0^2 \\
 &= 64\pi \left(\frac{1}{3} - \frac{1}{5} \right) - 0 \\
 \text{so } V &= \frac{128\pi}{15}.
 \end{aligned}$$

**3**

✓

✓

✓

Total for Question 11: 15 Marks**QUESTION TWELVE** (15 marks)**Marks**

(a)

For the hyperbola $5 = 4(e^2 - 1)$

so $e^2 = \frac{9}{4}$

thus $e = \frac{3}{2}$

1

✓

(b) Differentiate $x^2 - 2xy + 3y^2 = 11$ implicitly to get

$$2x - 2y - 2xy' + 6y'y = 0$$

or $(3y - x)y' = y - x$

so $y' = \frac{y - x}{3y - x}$

and at $(2, -1)$ $y' = \frac{3}{5}$

3

✓

✓

✓

(c) (i) Now $P'(z) = 4z^3 + 6z^2 + 6z + 4$, so

$$P'(-1) = 1 - 2 + 3 - 4 + 2$$

$$= 0$$

and $P(-1) = -4 + 6 - 6 + 4$

$$= 0$$

Hence $z = -1$ is a double root.**1**

✓

(ii) Let one root be α then $\bar{\alpha}$ is also a root (real coefficients)**3**

✓

product of roots: $\alpha \times \bar{\alpha} \times (-1) \times (-1) = 2$

so $|\alpha|^2 = 2$



sum of roots: $\alpha + \bar{\alpha} + (-1) + (-1) = -2$

so $\text{Re}(\alpha) = 0$

hence $\alpha = i\sqrt{2}$



and the roots are $(-1), (-1), i\sqrt{2}$ and $-i\sqrt{2}$.

(d)

3

$$v \frac{dv}{dx} = -\frac{1}{3}v^3 - v$$

so $\frac{dx}{dv} = \frac{-3}{v^2 + 3}$

Integrate this result to get

$$x = -\frac{3}{\sqrt{3}} \tan^{-1} \frac{v}{\sqrt{3}} + C$$



At $t = 0$, $x = 0$ and $v = 3$ so

$$0 = -\sqrt{3} \tan^{-1} \sqrt{3} + C$$

or $C = \frac{\pi}{\sqrt{3}}$



So $x = \frac{\pi}{\sqrt{3}} - \sqrt{3} \tan^{-1} \frac{v}{\sqrt{3}}$

thus $\tan^{-1} \frac{v}{\sqrt{3}} = \frac{\pi}{3} - \frac{x}{\sqrt{3}}$

hence $v = \sqrt{3} \tan \left(\frac{\pi}{3} - \frac{x}{\sqrt{3}} \right)$



(e) (i)

2

It should be clear that $r = \frac{3}{4}h$.



Thus Area = square – corners + circle

$$= 6 \times 6 - 4 \times r^2 + \pi r^2$$

$$= 36 + (\pi - 4)r^2$$



$$= 36 + (\pi - 4)\frac{9}{16}h^2$$

(ii)

2

Hence $V = \int_0^4 (36 + (\pi - 4)\frac{9}{16}h^2) dh$



$$= [36h + (\pi - 4)\frac{3}{16}h^3]_0^4$$

$$= 144 + 12(\pi - 4) - 0$$

$$= 12(8 + \pi)$$



Total for Question 12: 15 Marks

QUESTION THIRTEEN (15 marks)

Marks

- (a) (i) (α) In the following, write f for $f(x)$.

1

Here $g(x) = \frac{1}{f}$

so $\frac{dg}{dx} = -\frac{1}{f^2} \times f'$ (by the chain rule)

Thus $g'(a) = -\frac{f'(a)}{(f(a))^2}$
 $= 0$ (since $f'(a) = 0$)

✓

- (β) The product rule is easier, but most candidates used the quotient rule.

2

$g'' = -\frac{f^2 f'' - f' \times 2f' f}{f^4}$ (by the quotient rule)

$= \frac{2(f')^2 - f'' f}{f^3}$

✓

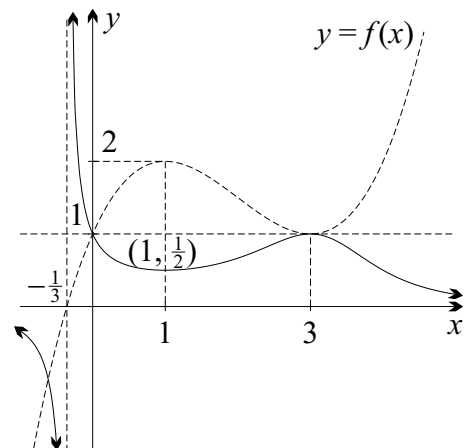
hence $g''(a) = \frac{0 - f''(a)f(a)}{(f(a))^3}$
 $= -\frac{f''(a)}{(f(a))^2}$

✓

The denominator is a square, so is positive,

hence $g''(a)$ has opposite sign to $f''(a)$.

- (ii)



2

Vertical and horizontal asymptotes

✓

Correct behaviour at turning points

✓

- (b) (i) In $\triangle APT$ and $\triangle SPT$

$\angle APT = \angle SPT$ (reflection property of ellipse)

2

$AP = PS$ (given)

✓

$PT = PT$ (common)

thus $\triangle APT \equiv \triangle SPT$ (SAS)

✓

hence $AT = ST$ (matching sides congruent triangles)

(ii) 1

$$S'A = S'P + PA$$

$$= S'P + PS \quad (\text{by construction})$$

$$= 2a \quad (\text{property of ellipse})$$

✓

(iii) (α) 2

Likewise, $S'B = 2a$ and $BT = ST$

✓

Hence in $\triangle S'AT$ and $\triangle S'BT$

$$S'T = S'T \quad (\text{common})$$

$$AT = BT = ST \quad (\text{proven})$$

$$S'A = S'B = 2a \quad (\text{proven})$$

thus $\triangle S'AT \equiv \triangle S'BT$ (SSS)

✓

Hence $\angle AS'T = \angle BS'T$ (matching angles of congruent triangles.)

(c) (i) 2

$$\angle BCP = \angle BAP \quad (\text{angles in the same segment})$$

$$= \alpha$$

✓

$$\angle QCA = \angle QBA \quad (\text{angles in the same segment})$$

$$= \beta$$

✓

Hence $\angle QCP = \alpha + \beta + \gamma$ (adjacent angles).

(ii) 2

$$2\alpha + 2\beta + \gamma = \pi \quad (\text{angle sum of } \triangle ABC)$$

so $\alpha + \beta = \frac{1}{2}(\pi - \gamma)$

✓

Thus $\angle QCP = \frac{1}{2}(\pi - \gamma) + \gamma$

$$= \frac{1}{2}(\pi + \gamma)$$

But γ is constant (angle in the major segment)

hence $\angle QCP = \frac{1}{2}(\pi + \gamma)$ is constant

✓

(iii) $\angle PCQ$ is constant hence 1

QP is also constant (equal angles subtend equal chords)

✓

Total for Question 13: 15 Marks

QUESTION FOURTEEN (15 marks)**Marks**

(a) Using the given results:

3

$$\begin{aligned}
 16 \cos^3 \theta \sin^2 \theta &= 16 \left(\frac{z + z^{-1}}{2} \right)^3 \left(\frac{z - z^{-1}}{2i} \right)^2 \\
 &= -\frac{1}{2} (z^3 + 3z + 3z^{-1} + z^{-3}) (z^2 - 2 + z^{-2}) \\
 &= -\frac{1}{2} (z^5 + z^3 - 2z - 2z^{-1} + z^{-3} + z^{-5}) \\
 &= -\frac{1}{2} (-2(z + z^{-1}) + (z^3 + z^{-3}) + (z^5 + z^{-5})) \\
 &= -\frac{1}{2} (-2 \cos \theta + 2 \cos 3\theta + 2 \cos 5\theta) \\
 &= 2 \cos \theta - \cos 3\theta - \cos 5\theta.
 \end{aligned}$$

✓

✓

✓

(b) (i) By de Moivre,

1

$$\operatorname{cis} 5\theta = \operatorname{cis} 2n\pi \quad \text{where } z = \operatorname{cis} \theta$$

$$\text{so} \quad 5\theta = 2n\pi$$

$$\text{or} \quad \theta = \frac{2n\pi}{5}.$$

$$\text{Hence} \quad z = 1, \operatorname{cis} \frac{2\pi}{5}, \overline{\operatorname{cis} \frac{2\pi}{5}}, \operatorname{cis} \frac{4\pi}{5}, \overline{\operatorname{cis} \frac{4\pi}{5}}$$

✓

(ii)

1

$$\alpha^5 - 1 = 0$$

$$\text{so} \quad (\alpha - 1)(\alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1) = 0 \quad (\text{GP theory})$$

$$\text{hence} \quad \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = 0 \quad (\text{since } \alpha \neq 1)$$

✓

(iii)

2

$$\text{Sum of roots:} \quad \alpha^4 + \alpha^3 + \alpha^2 + \alpha = -1 \quad (\text{from part (ii)})$$

✓

$$\begin{aligned}
 \text{Product of roots:} \quad (\alpha^4 + \alpha)(\alpha^3 + \alpha^2) &= \alpha^7 + \alpha^6 + \alpha^4 + \alpha^3 \\
 &= \alpha^2 + \alpha + \alpha^4 + \alpha^3 \quad (\text{since } \alpha^5 = 1) \\
 &= -1 \quad (\text{by part (ii)})
 \end{aligned}$$

$$\text{Hence a quadratic equation is } z^2 + z - 1 = 0.$$

✓

(iv)

2

$$\Delta = 5$$

$$\text{so} \quad z = \frac{-1 \pm \sqrt{5}}{2}$$

✓

But α is in the first quadrant so take the positive solution.

$$\text{Thus } \alpha^4 + \alpha = 2 \cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{2}$$

$$\text{hence } \cos \frac{2\pi}{5} = \frac{\sqrt{5} - 1}{4}$$

✓

- (c) (i) Since
- $P(\alpha) = 0$
- and
- $P(i\alpha) = 0$
- ,
- α
- is a solution of

2

$$z^4 - 2z^3 + 2z^2 - 10z + 25 = 0 \quad [1]$$

$$\text{and } z^4 + 2iz^3 - 2z^2 - 10iz + 25 = 0 \quad [2]$$

☒Subtracting [1] from [2], α is a solution of

$$(2i + 2)z^3 - 4z^2 - (10i - 10)z = 0$$

$$\text{or } 2z((1 + i)z^2 - 2z + 5(1 - i)) = 0$$

$$\text{thus } (1 + i)z^2 - 2z + 5(1 - i) = 0 \quad (\text{since } P(0) \neq 0)$$

☒

- (ii) The discriminant is

4

$$\Delta = 4 - 4 \times (1 + i) \times 5(1 - i)$$

$$= (6i)^2$$

☒

$$\text{so } z = \frac{2 \pm 6i}{2(1 + i)}$$

☒

$$= \frac{1 + 3i}{1 + i} \quad \text{or} \quad \frac{1 - 3i}{1 + i}$$

$$= \frac{4 + 2i}{2} \quad \text{or} \quad \frac{-2 - 4i}{2}$$

$$= 2 + i \quad \text{or} \quad -1 - 2i$$

☒The other roots are conjugates ($P(z)$ has real coefficients)

$$\text{thus } z = 2 + i, 2 - i, -1 + 2i, -1 - 2i$$

☒**Total for Question 14: 15 Marks****QUESTION FIFTEEN** (15 marks)**Marks**

- (a) There are two obvious methods to do this question. Here is the first.

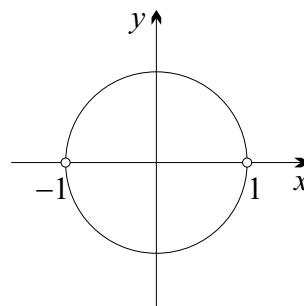
3

$$\frac{z - 1}{z + 1} = \frac{(z - 1)(\bar{z} + 1)}{|z + 1|^2}$$

$$= \frac{|z|^2 - 1 + z - \bar{z}}{|z + 1|^2}$$

$$= \frac{|z|^2 - 1}{|z + 1|^2} + \frac{2iy}{|z + 1|^2}$$

(or equivalent)

☒This is imaginary if $|z|^2 - 1 = 0$ ☒provided that $y \neq 0$.☒That is, a circle centre the origin and radius 1, omitting the x -intercepts.The other method is to note that $\arg\left(\frac{z - 1}{z + 1}\right) = \pm \frac{\pi}{2}$.

(b) (i) 1

$$\begin{aligned}
 (Q'Q)^2 &= (a\mu - a\lambda)^2 + (b\mu - b\lambda)^2 \\
 &= a^2(\mu - \lambda)^2 + b^2(\mu - \lambda)^2 \\
 &= (\mu - \lambda)^2(a^2 + b^2)
 \end{aligned}$$

✓

so $Q'Q = (\mu - \lambda)\sqrt{a^2 + b^2}$ (since $\mu > \lambda$).

(ii) In $\triangle AFP \parallel \triangle AQR$ 1

$$\frac{QR}{AR} = \frac{PF}{AF} \quad (\text{matching sides of similar triangles})$$

so $\frac{b\mu}{a\mu - a} = \frac{b \tan \theta}{a \sec \theta - a}$

thus $\frac{\mu}{\mu - 1} = \frac{\tan \theta}{\sec \theta - 1}$

✓(iii) In $\triangle A'FP \parallel \triangle A'Q'R'$ (AA) 1

$$\frac{Q'R'}{A'R'} = \frac{PF}{A'F} \quad (\text{matching sides of similar triangles})$$

so $\frac{b\lambda}{a\lambda + a} = \frac{b \tan \theta}{a \sec \theta + a}$

thus $\frac{\lambda}{\lambda + 1} = \frac{\tan \theta}{\sec \theta + 1}$

✓(iv) 2

$$\begin{aligned}
 \frac{\mu}{\mu - 1} \times \frac{\lambda}{\lambda + 1} &= \frac{\tan \theta}{\sec \theta - 1} \times \frac{\tan \theta}{\sec \theta + 1} \\
 &= \frac{\tan^2 \theta}{\sec^2 \theta - 1} \\
 &= 1
 \end{aligned}$$

✓

thus $\lambda\mu = (\mu - 1)(\lambda + 1)$

or $\lambda\mu = \lambda\mu + \mu - \lambda - 1$

so $\mu - \lambda = 1$

hence $Q'Q = \sqrt{a^2 + b^2}$

✓

which is independent of the location of P on $\mathcal{H}$.

(c) Squares of real numbers are positive, so 1

$$\left(x - \frac{1}{x}\right)^2 \geq 0$$

thus $x^2 + \frac{1}{x^2} \geq 2$

Now put $a = x^2$ to get the result

$$a + \frac{1}{a} \geq 2$$

✓

- (d) (i) Rearrange the given equation to get

3

$$\frac{dt}{dv} = \frac{-1}{g + kv}$$

thus $t = -\frac{1}{k} \log(g + kv) + C$ (for some constant C .)

✓

At $t = 0$, $v = V_0$ so

$$C = \frac{1}{k} \log(g + kV_0)$$

✓

Thus $t = \frac{1}{k} \log(g + kV_0) - \frac{1}{k} \log(g + kv)$

or $e^{kt} = \frac{g + kV_0}{g + kv}$

so $g + kv = g(1 + \frac{k}{g}V_0)e^{-kt}$

hence $v(t) = \frac{g}{k} \left((1 + \frac{k}{g}V_0)e^{-kt} - 1 \right)$

✓

- (ii) At the maximum height the velocity is zero, so

1

$$(1 + \frac{k}{g}V_0)e^{-kT} - 1 = 0$$

or $1 = (1 + \frac{k}{g}V_0)e^{-kT}$

hence $e^{kT} = (1 + \frac{k}{g}V_0)$

✓

- (iii)

2

$$v(T - s) + v(T + S) = \frac{g}{k} \left((1 + \frac{k}{g}V_0)e^{-kT+ks} - 1 \right) + \frac{g}{k} \left((1 + \frac{k}{g}V_0)e^{-kT-ks} - 1 \right)$$

$$= \frac{g}{k} \left((e^{kT}e^{-kT+ks} - 1) + (e^{kT}e^{-kT-ks} - 1) \right) \quad (\text{part (ii)})$$

✓

$$= \frac{g}{k} (e^{ks} + e^{-ks} - 2)$$

$$= \frac{g}{k} \left(a + \frac{1}{a} - 2 \right) \quad (\text{where } a = e^{ks})$$

$$\geq \frac{g}{k} (2 - 2) \quad (\text{by part (c)})$$

✓

$$\geq 0.$$

Total for Question 15: 15 Marks

QUESTION SIXTEEN (15 marks)

Marks

- (a) (i) The given equation will hold if

2

$$1 = A(x^2 + 1) + (Bx + C)(x + 1)$$

at $x = -1$ this gives

$$1 = 2A \quad \text{so} \quad A = \frac{1}{2}$$

✓

equating coefficients of x^2 gives

$$0 = A + B \quad \text{so} \quad B = -\frac{1}{2}$$

and at $x = 0$ the result is

$$1 = A + C \quad \text{so} \quad C = \frac{1}{2}$$

✓

[Only full marks if all three correct.]

(ii) From the results of part (i),

3

$$\begin{aligned}
I(N) &= \frac{1}{2} \int_0^N \frac{1}{x+1} - \frac{x}{x^2+1} + \frac{1}{x^2+1} dx \\
&= \frac{1}{2} \left[\log(x+1) - \frac{1}{2} \log(x^2+1) + \tan^{-1} x \right]_0^N \\
&= \frac{1}{2} \left(\log(N+1) - \frac{1}{2} \log(N^2+1) + \tan^{-1} N \right) \\
&= \frac{1}{4} \left(\log \left(\frac{(N+1)^2}{N^2+1} \right) + 2 \tan^{-1} N \right)
\end{aligned}$$

 $\checkmark\checkmark$ $\checkmark$ (iii) Taking the limit of part (ii) as $N \rightarrow \infty$:**1**

$$\begin{aligned}
\int_0^\infty \frac{dx}{x^3+x^2+x+1} &= \lim_{N \rightarrow \infty} I(N) \\
&= \lim_{N \rightarrow \infty} \frac{1}{4} \left(\log \left(\frac{(N+1)^2}{N^2+1} \right) + 2 \tan^{-1} N \right) \\
&= \lim_{N \rightarrow \infty} \frac{1}{4} \left(\log \left(\frac{(1+\frac{1}{N})^2}{1+\frac{1}{N^2}} \right) + 2 \tan^{-1} N \right) \\
&= \frac{1}{4} (\log 1 + 2 \times \frac{\pi}{2}) \\
&= \frac{\pi}{4}
\end{aligned}$$

 $\checkmark$ (b) (i) Using integration by parts $\int v' u dx = uv - \int u' v dx$,**1**

$$\begin{aligned}
&\text{put } u = e^{-x} \quad \text{and } v' = x^n \\
&\text{so } u' = -e^{-x} \quad \text{and } v = \frac{1}{n+1} x^{n+1} \\
&\text{then } I_n = \left[\frac{x^{n+1} e^{-x}}{n+1} \right]_0^1 + \int_0^1 \frac{x^{n+1} e^{-x}}{n+1} dx \\
&= \frac{e^{-1} - 0}{n+1} + \frac{1}{n+1} \int_0^1 x^{n+1} e^{-x} dx \\
&= \frac{1}{n+1} (e^{-1} + I_{n+1})
\end{aligned}$$

 $\checkmark$ (ii) (A) When $n = 1$ **3**

$$\begin{aligned}
\text{LHS} &= \frac{1}{1} (e^{-1} + I_1) \quad (\text{by part (i)}) \\
&= e^{-1} \times \frac{1}{1!} + \frac{1}{1!} \times I_1 \\
&= \text{RHS}
\end{aligned}$$

 $\checkmark$ so the result is true for $n = 1$.(B) Assume the result is true when $n = k$. That is, assume that

$$I_0 = e^{-1} \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} \right) + \frac{1}{k!} I_k \quad (\dagger\dagger)$$

Now prove the result is true when $n = k + 1$. That is, prove that

$$I_0 = e^{-1} \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} + \frac{1}{(k+1)!} \right) + \frac{1}{(k+1)!} I_{k+1}$$

Now, by part (i), the result in (††) becomes

$$I_0 = e^{-1} \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} \right) + \frac{1}{k!} \times \frac{1}{k+1} (e^{-1} + I_{k+1}) \quad \checkmark$$

$$= e^{-1} \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} \right) + \frac{e^{-1}}{(k+1)!} + \frac{1}{(k+1)!} I_{k+1}$$

$$= e^{-1} \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{k!} + \frac{1}{(k+1)!} \right) + \frac{1}{(k+1)!} I_{k+1} \quad \checkmark$$

as required.

It follows from parts (A) and (B) by mathematical induction that the result is true for all positive integers n .

(iii) Take the limit as $n \rightarrow \infty$ as indicated to get

1

$$I_0 = e^{-1} \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots \right)$$

but
$$I_0 = \int_0^1 e^{-x} dx$$

$$= \left[e^{-x} \right]_0^1$$

$$= -e^{-1} + 1$$

Hence $1 - e^{-1} = e^{-1} \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots \right)$ $\checkmark$

so $e - 1 = \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots$

or $e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots$

(iv) Let $e = \frac{p}{q}$ then

1

$$\frac{p}{q} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{q!} + \frac{1}{(q+1)!} + \frac{1}{(q+2)!} + \dots$$

Now multiply both sides by $q!$ to get

$$\frac{p q!}{q} = \underbrace{q! + \frac{q!}{1!} + \frac{q!}{2!} + \frac{q!}{3!} + \dots + \frac{q!}{q!}}_{\left(\sum_{k=0}^q \frac{q!}{k!} \right)} + \frac{q!}{(q+1)!} + \frac{q!}{(q+2)!} + \dots \quad \checkmark$$

or $p(q-1)! = \left(\sum_{k=0}^q \frac{q!}{k!} \right) + \frac{1}{q+1} + \frac{1}{(q+1)(q+2)} + \dots$

(v) Let $f = \frac{1}{q+1} + \frac{1}{(q+1)(q+2)} + \frac{1}{(q+1)(q+2)(q+3)} + \dots$ then **2**

$$f < \frac{1}{q+1} + \frac{1}{(q+1)^2} + \frac{1}{(q+1)^3} + \dots \quad (\text{decrease denominators}) \quad \checkmark$$

$$< \frac{1}{q+1} \times \frac{1}{1 - \frac{1}{q+1}} \quad (\text{infinite sum of a GP})$$

$$< \frac{1}{q}$$

$$< \frac{1}{2} \quad (\text{since } q > 2) \quad \checkmark$$

Also $f > 0$ since every term is positive. Hence $0 < f < \frac{1}{2}$.

(vi) The LHS of part (iv) is an integer. **1**

The RHS of part (iv) is an integer plus a fraction.

This is a contradiction and hence e is a irrational. **✓**

Total for Question 16: 15 Marks

DNW